Problem 2: What is the last digit of $4^{3^{2019}}$?

Solution:

We can write a table that gives the value of 4^n for all positive integers n up to n=20. Then we observe a pattern: 4^n ends in 4 when n is odd, and 4^n ends in 6 when n is even.

Since 3^{2019} is an odd number, we can guess that the last digit of $4^{3^{2019}}$ is a 4.

Now, let's prove it.

Let n be a positive integer that has 4 in its ones digit (or equivalently, n is equal to 4 mod 10). We can write $n = 10k + 4$ by finding a suitable integer value for k. If we multiply n by 4, we get

$$\begin{aligned} 4n &= 4(10k + 4) \\ &= 40k + 16 \\ &= 40k + 10 + 6 \\ &= 10(4k + 1) + 6 \end{aligned}$$

We can infer that $4n$ has a 6 in its ones digit, because $10(4k + 1) + 6$ necessarily ends in 6.

Let m be a positive integer that has 6 in its ones digit (or equivalently, m is equal to 6 mod 10). We can write $m = 10j + 6$ by finding a suitable integer value for j. If we multiply m by 4, we get

$$\begin{aligned} 4m &= 4(10j + 6) \\ &= 40j + 24 \\ &= 40j + 20 + 4 \\ &= 10(4j + 2) + 4 \end{aligned}$$

We can infer that $4m$ has a 4 in its ones digit, because $10(4j + 2) + 4$ necessarily ends in 4.

Using these conclusions, we can show, by induction, that 4^n ends in a 4 if n is odd, and 4^n ends in a 6 if n is even. We assume the rule is true for n; then, our work beforehand shows that it's true for $n + 1$. It's true for the base case, where 4^1 has a 4 in its ones digit. Thus it's true for all positive integers n.

Since $4^{3^{2019}}$ is four raised to the power of an odd number, we can deduce that the last digit of $4^{3^{2019}}$ is a four.